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Article in IET Nanobiotechnology · August 2015

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Development of shampoo, soap and ointment formulated by green synthesised silver nanoparticles functionalised with antimicrobial plants oils in veterinary dermatology: treatment and prevention strategies

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ISSN 1751-8741

Received on 12th August 2014

Accepted on 13th November 2014

doi: 10.1049/iet-nbt.2014.0042

www.ietdl.org

Abstract: Many scientists have focused their research on the role of nanotechnology for the control of human pathogens, but there are also many topical pathogens present in animals, which infect animals and transfer to humans. Topical therapy is extremely important for the management of dermatological condition in animals. Therefore, the present study aims to evaluate the efficacy of biogenic silver nanoparticles (AgNPs) in combination with herbal oils against animal skin infections which may be responsible for causing infections in human beings. Here, the authors synthesised and characterised the AgNPs from *Azadirachta indica*. The oils were extracted from medicinal plants including *Cymbopogon citratus*, *Cymbopogon martini*, *Eucalyptus globules*, *A. indica* and *Ocimum sanctum* and the antifungal and antibacterial activity of plant oils along with AgNPs were evaluated. An excision wound model was used for the study of wound healing activity in rabbits. AgNPs functionalised oil has demonstrated remarkable antimicrobial activity against pathogens present on the skin of animals. The nano-functionalised antimicrobial oils were used in the formulation of shampoo, soap and ointment for veterinary dermatology. Antimicrobial products of plant origin with AgNPs are valuable, safe and have a specific role in controlling diseases. The authors believe that this approach will be a good alternative therapy to solve the continuous antibiotic resistance developed by many bacterial pathogens and will be utilised in various animal contacting areas in medicine.

1 Introduction

Nanotechnology is a multidisciplinary subject dealing with the synthesis, properties and applications of the particles at the nano range (1–100 nm). Researchers have been developing various products based on nanoparticles, some of which are open to the market for public use. Nanotechnologies have great effect on house-cleaning, antiseptics and disinfectant products. Nanoparticles are also used in cleaning fluids which have been engineered to absorb dust and grime on contact [1, 2]. The use of plants in the synthesis of gold nanoparticles has emerged as an eco-friendly and exciting approach [3–5]. There are many reports on the synthesis of silver nanoparticles (AgNPs) using plant extracts such as *Olea europaea* leaves [6], *Eucalyptus chapmaniana* [7] and *Saraca indica* bark [8].

In humans or higher animals, fungal and bacterial diseases are mostly mentioned in relation to infections of superficial and subcutaneous organs, especially skin and other peripheral organs. With the emergence and increase of microbial resistance to multiple antibiotics, and the continuing emphasis on healthcare costs, there is a need to develop a cost-effective and resistance-free antimicrobial agents. Skin infections are the most common in animals that mainly suffer from bacterial, yeast, fungal and external parasitic infection. Itchiness, hair loss, red skin, small bumps in the skin, pimple-like lesions and scaling are common symptoms in most infections.

There are a vast number of skin infections in pets. Some of the common dermatological conditions include flea allergy dermatitis, atopy (environmental allergy), otitis (ear disease), bacterial diseases (staph infection), parasitic skin diseases, fungal skin diseases (such as ringworm), food allergy, skin cancer, skin

diseases of the foot or nail, seborrhea and acne, alopecia (hair loss), nutritional skin diseases, endocrine and metabolic skin diseases, auto-immune skin diseases and drug reactions. In fact, many pets suffer from more than one dermatological infection at a time. In farm animals, antimicrobial agents are also used to enhance performance by increasing feed conversion, growth rate or yield. Prophylaxis or strategic treatment is normally used to contain the spread of infection and prevent illness before the development of clinical signs. In short, there is a need for antimicrobial drugs in veterinary medicine to treat animal pathogens [9]. Antimicrobial drugs are given to animals by injection, orally in food or water topically on the skin. We know that bacteria and fungi are resistant to some antibiotics and antifungal drugs. Therefore, we need to develop novel antibiotic and antimicrobial products such as antimicrobial soap, cream, iodine tincture and shampoo for animals. Ayurvedic medicinal systems use noble metals, mainly gold and silver [10]. The shampoos are used not only for cleansing purpose, but also for the removal of bacteria and fungi present on skin of animals and to maintain the manageability and oiliness of hair. Shampoo is generally used to remove dirt on hair. Microorganisms, plant extract and biomass thus could serve as alternative to chemical and physical methods for the production of AgNPs in an eco-friendly manner. AgNPs have the properties of high surface area, very small size (<20 nm) and high dispersion [11]. Silver is a safe and effective bactericidal metal because it is non-toxic to animal cells and highly toxic to bacteria [12–14].

In our paper, we synthesised AgNPs from plant material, which showed antimicrobial activity. These AgNPs were functionalised with herbal oils (surface modified AgNPs with herbal oil) to develop

antimicrobial products for animal skin diseases. Nanoparticles, being nanosized, exhibit smaller volume to greater surface area ratio and have high dispersion property; hence, they are utilised for the formulation of nano-functionalised antimicrobial product. AgNPs functionalised with mixed plant oils showed a higher antimicrobial activity compared with plant oil alone or AgNPs. This nano-herbal product is without side effects and cost-effective. The aims of the current paper were (i) to use the extract of *Azadirachta indica* for the synthesis of AgNPs, (ii) to evaluate the antibacterial activity of the plant extract alone and with surface modified AgNPs and (iii) to develop antimicrobial shampoo, soap and ointment for the treatment and prevention strategies of skin diseases in animals.

2 Materials and methods

2.1 Green synthesis of AgNPs from *A. indica*

The phytosynthesis was carried out by using the leaf extract of *A. indica*. Fresh leaves of the plant were collected and washed with 70% alcohol for 10 min, followed by washing with autoclaved water. The leaves were finely cut, boiled in 100 ml of distilled water and filtered through Whatman filter paper No. 1. The solution was centrifuged at 4000 rpm and filtered using a vacuum filter. The plant extract was prepared by mixing 1% leaf extract with distilled water in a 250 ml Erlenmeyer flask. It was then challenged with 1 mM silver nitrate (AgNO_3) for the synthesis of AgNPs. The colour change indicates the formation of AgNPs (Fig. 1 inset).

2.2 Detection of AgNPs synthesised from *A. indica*

The primary detection was based on observing the colour change within an hour of incubation. The synthesis of nanoparticles was confirmed by measuring absorbance using an ultraviolet–visible (UV–vis) spectrophotometer (Shimadzu UV-1700, Japan). Samples of biogenic AgNPs were scanned in the range of wavelength 200–800 nm at a resolution of 1 nm (Fig. 1).

2.3 Purification of synthesised AgNPs

The synthesised AgNPs were then purified and concentrated by repeated centrifugation at 15 000g for 30 min. The supernatant was discarded and the pellets were dissolved in deionised water. These concentrated nanoparticle samples were used for further characterisation. The AgNPs were characterised by using Fourier

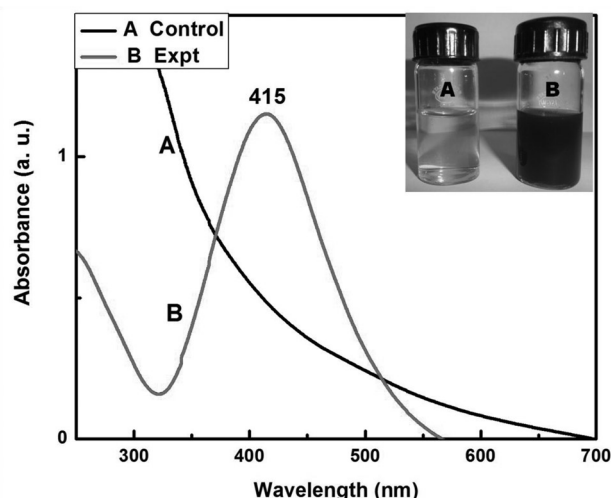


Fig. 1 UV–vis spectrum of the AgNPs synthesised by *A. indica* leaf extract and inset showing colour change of *Azadirachta* leaf extract

a Before treatment of 1 mM AgNO_3
b After treatment of 1 mM AgNO_3

transform infrared spectroscopy (FTIR), LM20, zeta potential measurement and transmission electron microscopy (TEM) analysis.

2.4 TEM analysis

5 μl of the sample was taken for the characterisation by TEM and three images of each sample were recorded to have a clear representation of its topology (Fig. 2).

2.5 Nanoparticle tracking and analysis (NTA) system (LM20)

Nanosight (LM20, UK) was used to measure the size of synthesised AgNPs. The sample was diluted with the sterile water and 5 μl of sample was injected onto the sample chamber and observed through the LM20 to determine the size of the AgNPs (Fig. 3).

2.6 FTIR analysis

AgNPs were characterised by FTIR (Perkin-Elmer FTIR-1600, USA) in the range of 400–4000 cm^{-1} at a resolution of 4 cm^{-1} . Samples were dried with potassium bromide in a hot air oven, crushed in a mortar and pestle and subjected to FTIR analysis (Fig. 4).

2.7 Medicinal plant oils

The leaves of *Cymbopogon citratus* Stapf., *Cymbopogon martinii* Roxb., *Eucalyptus globulus* Labill., *A. indica* Linn. and *Ocimum sanctum* Linn. were collected from the wild fields of Amravati University, Amravati district of Maharashtra, India and cut into pieces for extraction. *Eugenia caryophyllata* Thunb. oil was purchased from the local market of Amravati.

2.8 Extraction procedure

The air-dried aerial parts of leaves (50 g) were hydro distilled in a clevenger apparatus (Sigma Corp.) for 5 h in accordance with the British pharmacopoeia. The yield was 0.62% of dry weight. The aqueous phase was extracted with dichloromethane (Qualigens) (3 $\times$ 50 ml). The organic phase was dried with sodium sulphate (Bio-Rad Laboratories), filtered and the solvent was evaporated until dry by air-drying. The fractions obtained were combined into calibrated flasks, evaporated until dry and weighed in order to determine the extraction's efficiency. The oils were solubilised in dimethyl sulphoxide (DMSO) (Bio-Rad Laboratories) to a final

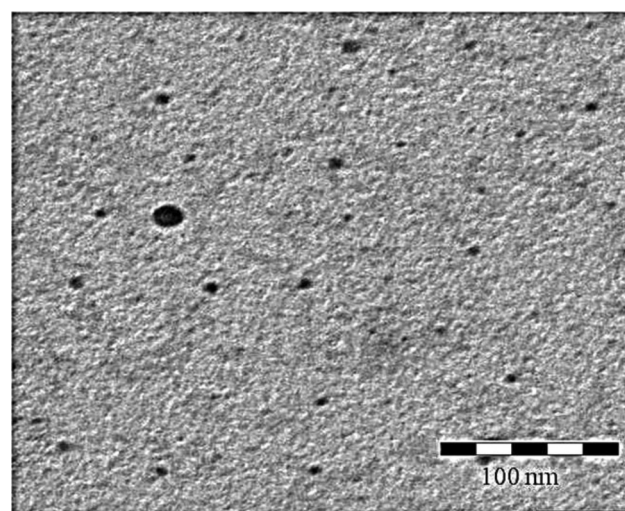


Fig. 2 TEM analysis of AgNPs revealed that the size of the nanoparticles ranged between 15 and 35 nm

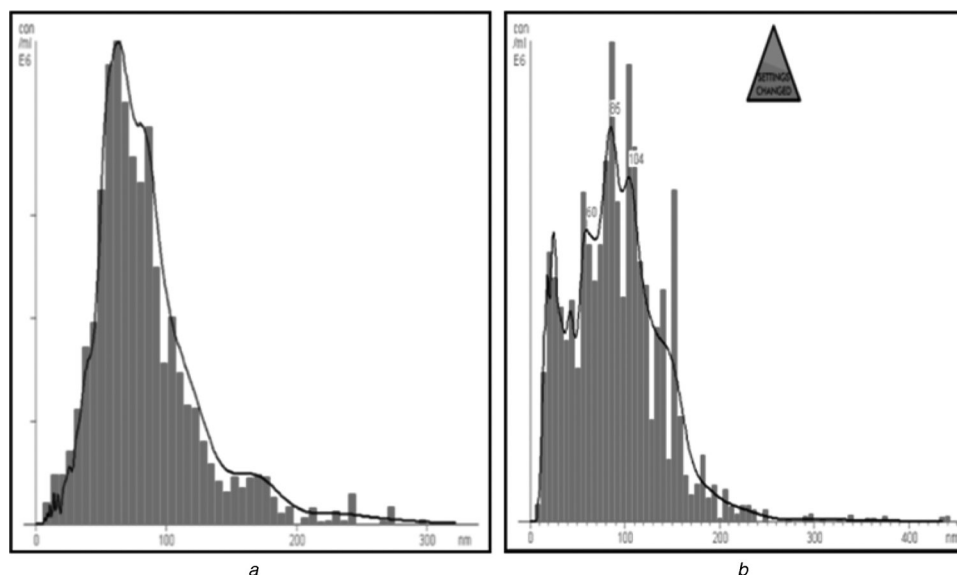


Fig. 3 NTA system (LM20)

a LM analysis of AgNPs revealed that the average size of the nanoparticles is 63 nm

b LM analysis of AgNPs conjugate with antimicrobial plant oils/extract revealed that the average size of the nanoparticles is 85 nm

concentration 5 mg/ml [15] and stored in a sealed glass vial (Bijou bottle) in a refrigerator at 4°C for further use. These oils were screened for their antibacterial and antimycotic activity.

2.9 Antifungal and antibacterial activity of plant oils

Plant oils were screened for their antifungal and antibacterial activity against *Aspergillus niger* isolated from animal skin and *Staphylococcus aureus* (ATCC 14948), *Escherichia coli* (ATCC 33591), by the disc diffusion method. Different concentrations of each oil and mixed oils (equal ratio of each oil), that is, 100, 50, 25 and 12.5 µg/disc, were used for assay. The fungal cultures were grown on Czapek dox broth (diffco) and bacteria were grown on nutrient agar. The mycelial mat of *A. niger* of 7-day-old culture was washed, suspended in normal saline solution and then filtered through glass wool aseptically. The colony forming units (CFUs/ml) of the fungal suspension were determined and test inoculum was adjusted to $1-5 \times 10^5$ ml. Fungal inoculum (0.1 ml) was applied onto the surface of the Czapek dox agar (Diffco) plate and spread by using a sterile glass spreader. The sterile discs (5 mm diameter, Whatman filter paper No. 42) were soaked in different concentrations (100, 50, 25 and 12.5 µg/disc) of essential oils. The test was performed in triplicate. These plates were incubated for 48 h at 28°C. The zone of inhibition in millimetres (mm) was measured after 48 h. Standard antibiotics miconazole (Sigma) and

itraconazole (Sigma) were used as a control to assess the sensitivity of the test fungus. DMSO (Sigma) and a filter paper disc soaked in distilled water without oil were used as negative control. The zones of different oils were measured. The data of all

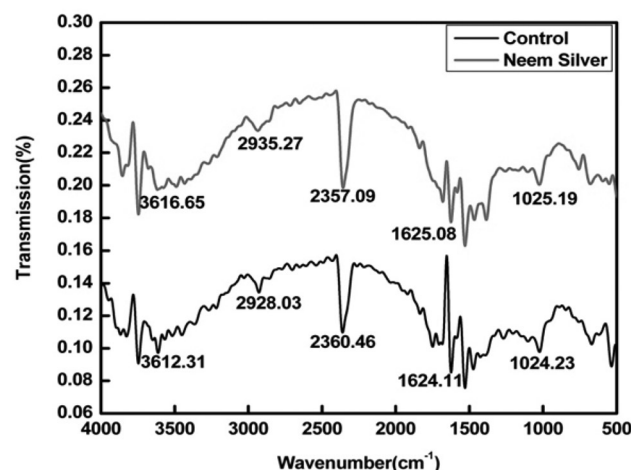


Fig. 5 Characterisation of synthesised AgNPs by FTIR

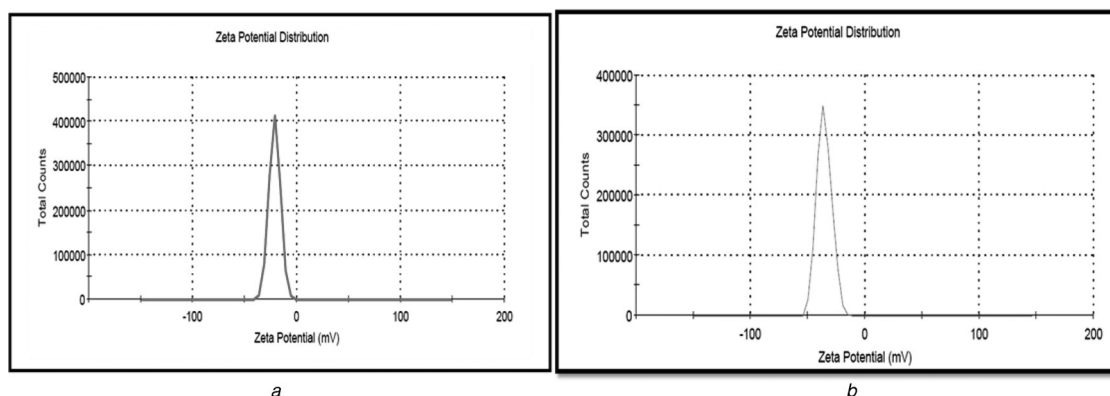


Fig. 4 Zeta potential was measured by Malvern Zetasizer 90 (ZS 90, USA) to determine the stability of NPs

a Zeta potential of AgNPs

b Zeta potential of shampoo containing plant oil conjugate AgNPs

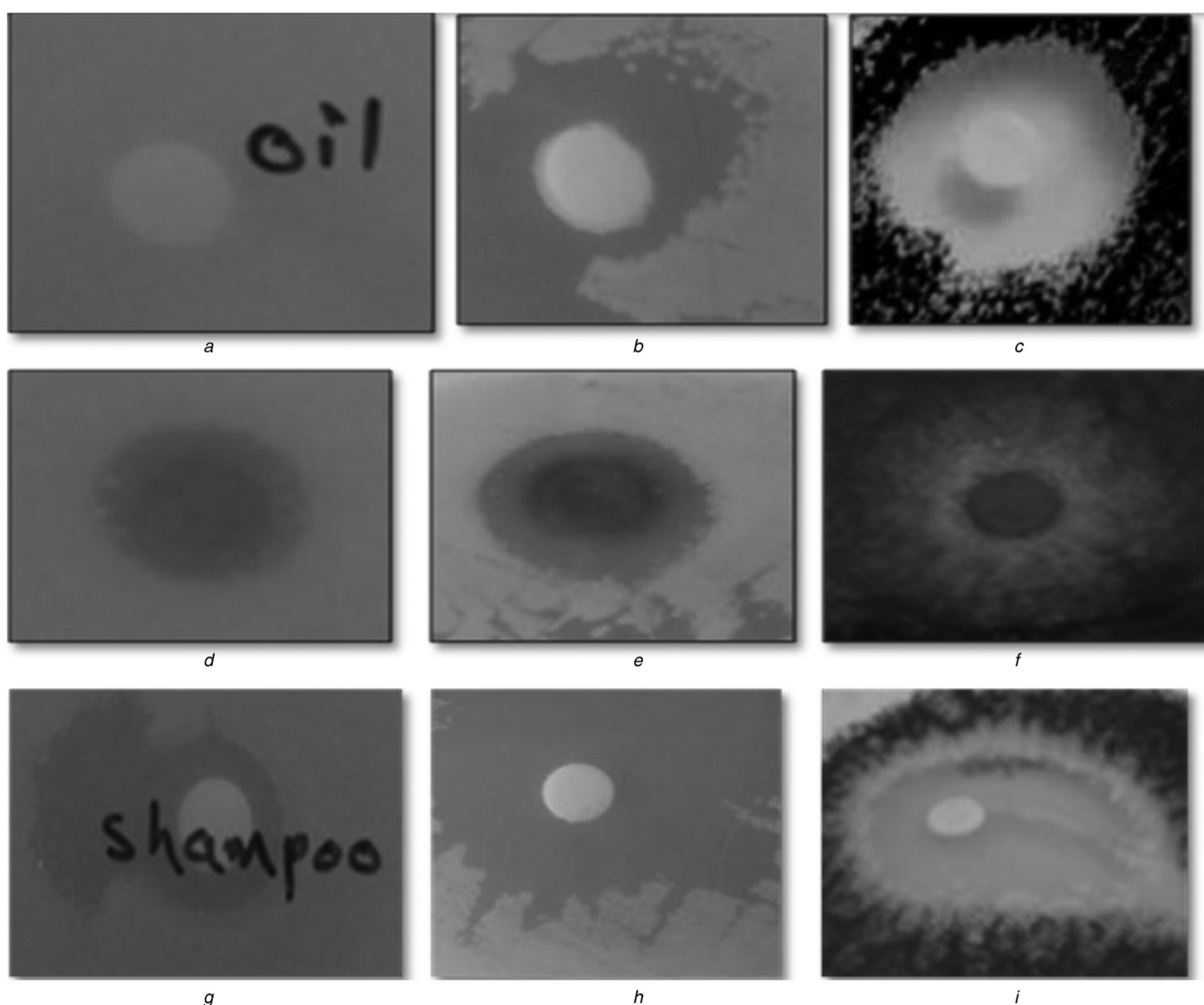


Fig. 6 Leaf oil and AgNPs alone exhibit less antimicrobial effect against *S. aureus*, *E. coli*, *A. niger* and *A. fumigatus*

a, b, c Antimicrobial activity of mixed plant oils

d, e, f Antimicrobial activity of AgNPs

g, h, i Antimicrobial activity of mixed antimicrobial oil conjugate with biosynthesised AgNPs using the test microbes *S. aureus*, *E. coli* and *A. niger*

the parameters were statistically analysed. The same concentration of each oil and mixed oils were used to evaluate the activity against *S. aureus* and *E. coli*, grown on nutrient agar by disc diffusion method. These plates were incubated at 37°C for 24 h.

2.9.1 Functionalised AgNPs with antimicrobial plant oils:

AgNPs (1 mg/ml) solution was emulsified with 1 ml of plant oil and mixed to form a homogeneous emulsion and incubated at room temperature for 30 min. The emulsion was characterised by NTA system (LM20) and the size and stability of NPs were also determined (Figs. 3 and 4).

2.9.2 Antimicrobial activity of AgNPs functionalised by oil:

The common human and animal pathogenic bacterial strains *E. coli* (ATCC 14948) and *S. aureus* (ATCC 33591) were used for assessment of antibacterial activity by the Kirby–Bauer disc diffusion method. The pathogenic fungi of animal origin *A. niger* and *Aspergillus fumigatus* were used for evaluation of antifungal activity of green synthesised AgNPs. For the disc diffusion method, log phase bacterial inocula (10^6 CFU/ml) were standardised against MacFarland's standard and were swabbed onto nutrient agar plates. Filter paper discs saturated with AgNPs conjugate with plant oil and extract were placed onto the surface of the medium with the help of sterile forceps and incubated at 37°C. After 18 h of incubation, the plates were examined for evidence of zone of inhibition, which appears as a clear area around the discs. All the experiments were carried out in triplicate. A similar protocol was used for assessment

of fungi and the medium used was Czapek's dox agar, incubated for 48 h at 28°C (Fig. 6).

2.10 AgNPs functionalised with herbal oils for soap and shampoo preparation

The AgNPs-plant oils emulsion was prepared by mixing the biogenic nanoparticles with natural plant oils. Later, this emulsion was used for the preparation of soap and shampoo. AgNPs offer unique properties such as small size, large surface area, high drug loading capacity and the interaction of phases at the interfaces and are attractive for their potential to improve performance of pharmaceuticals, nutraceuticals and other material [16].

2.11 Development of soaps for animals from AgNPs functionalised with antimicrobial herbal oils

About 5 ml of surface modified AgNPs with oils were taken in a beaker, and then 15 ml of ethanol and 15 ml of 20% sodium hydroxide were added to the beaker. The mixture was heated and stirred by a magnetic stirrer. The mixture (with constant stirring) was heated for 30 min until the solution developed two separate layers. The solution was transparent at this point. On complete saponification, the beaker was carefully removed from the heat. The soap solution was then mixed with 50 ml of a saturated sodium chloride (NaCl) (salt) solution in a 400 ml beaker and

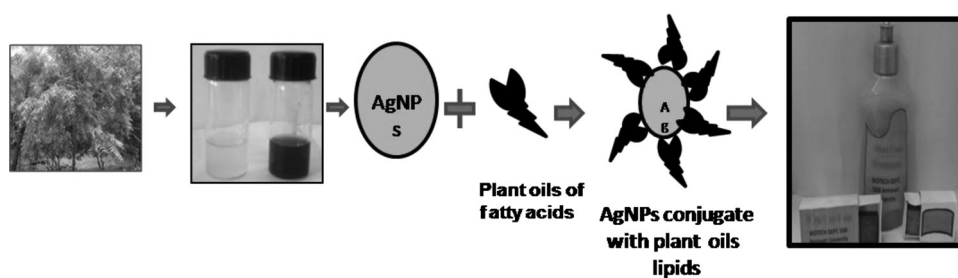


Fig. 7 Graphical abstract: developed animals soap and shampoo by using plant oil conjugate with AgNPs

stirred with a stirring rod. This 'salting out' process increases the density of the solution and causes the soap to precipitate and float on the surface of the solution. The beaker was placed in an ice-water bath until it reached the approximate temperature of the bath. About 20 ml of chilled deionised water was also kept in a separate container. Perfume and other ingredients were added as additives before pouring the saponification mixture into moulds. After pouring, the soap was allowed to harden by air-drying for 24 h to obtain the soap bars, according to the method reported by Warra *et al.* [16] and the soap bars were then observed for colour, texture, lathering and cleaning power. The pH test, foam test, interaction with oil and hard water test properties of herbal nanosoap were then tested and compared with a commercial soap. For the pH test, soap solutions were taken in four different test tubes. In the first tube was placed 10 ml of the prepared soap solution, in the second tube, 10 ml of the commercial soap solution, in the third tube, 10 ml of the detergent solution and in the fourth tube, 10 ml of deionised water (the control). One by one, each solution was stirred with a stirring rod and the pH of each solution was recorded. These solutions were then used for the foam test. Each of the tubes for the pH test was stoppered and continuously shaken for 10 s. The amount of foam formed was recorded. For interaction with oil, five drops of oil were added to each test tube for foam test. Each of the tubes were continuously stoppered and shaken well for 10 s. These were then observed for formation of oil layer in each tube. The amount of suds formed in each tube was compared with the amount of suds they each had in part of foam test. For the hard water test, 5 ml of the soap solution in first tube, 5 ml of commercial soap solution in the second tube and 5 ml of the detergent solution in the third tube were taken. About 20 drops of 1% calcium chloride (CaCl_2) solution were added to each tube and were stoppered and shaken continuously for 10 s. The amount of suds formed in each tube was compared with the amount of suds they each had in foam test. Observations about solutions were recorded. New samples of each soap solution were taken but instead of adding 1% CaCl_2 solution, 20 drops of 1% MgCl_2 solution were added. Again, the amount of foam produced was observed and compared with the amount produced in foam test. Again, new samples of each soap solution were taken with addition of 20 drops of 1% ferric chloride solution and observations were recorded.

2.12 Preparation of shampoo

Shampoo contains some specific functions along with the cleansing action. Shampoo usually contains medicinal agents such as vitamins, amino acid, plant extract, antibacterial and antifungal agent etc. [17]. The pH of the skin is found to be 6.8. The viscosity of shampoo is a more important factor because of spreadability on hairs. First, antimicrobial oil (1 ml) was mixed with AgNPs (1 mg/ml), carboxy methyl cellulose (2% gel) and a solution in a non-leaching clean glass or stainless steel container was made. Ethylenediaminetetraacetate was mixed in a small amount of water and then oil was added with AgNPs colloid. Then a small amount of saturated solution of NaCl was added dropwise into it. NaCl acts as viscosity modifier by the common ion effect. Colour and perfume were added and the shampoo was packed in a suitable container for evaluation. Then, herbal shampoo containing antimicrobial oils conjugated AgNPs was evaluated. The appearance of the prepared

emulsion was visually observed as a brownish cream. The viscosity of the prepared shampoo was determined by viscometer, pH test was performed by using the standard pH paper [18, 19] and foam test was determined by cylinder shake method [20, 21].

2.13 Preparation of ointment

According to topical formulation, emulsifying ointment base was prepared using 60% polyethylene glycol 4000. To this base, plant oil surface modified AgNPs were mixed in a beaker. It was heated in a 60°C water bath.

2.14 Wound healing activity

A rabbit already having wound was used as an experimental model for studying wound healing activity. An area of about 2 cm was defined with marker on the wound of the ear of the rabbit. All the samples, Control (received no treatment), Standard (streptomycin) and Test (AgNPs formulated cream), were applied once daily for 15 days starting from the day of wounding. The observations of the percentage of wound the closure were made on the 4th, 8th, 12th and 16th post-wounding days.

Wound contraction % = $\frac{\text{difference in the area of the wound in } \text{m}^2 \text{ between the initial and on a particular post-operative day}}{100/\text{area of the wound in mm immediately after the wound excision}} \times 100$ [22] (Fig. 7).

3 Results

In the present study, the results of the synthesis and characterisation of AgNPs revealed that treatment of *A. indica* extract with 1 mM aqueous AgNO_3 resulted in the development of the brown solution indicating the rapid biosynthesis of AgNPs (Fig. 1 inset). Spectrophotometric scanning of the produced brown coloured solution through the spectral range of 200–800 nm showed a maximum absorption at 415 nm (Fig. 1). The TEM analysis revealed that the size of AgNPs ranged between 15 and 35 nm (Fig. 2). Synthesised AgNPs were also analysed by LM20, which revealed the size of AgNPs as 63 nm (Fig. 3) with a zeta potential of -31.2 indicating the stability of the nanoparticles (Fig. 4). FTIR analysis shows the shifting in the absorption peaks at 2935 cm^{-1} (C–H stretch), 1625 cm^{-1} (N–H bend) and 1025 cm^{-1} (C–N stretch) corresponding to the functional groups that confirm the presence of capping proteins on the AgNPs stabilising the synthesised nanoparticles (Fig. 5). We obtained antimicrobial oils from plant origin; here, we found maximum activity of *C. martinii* (tikhadi oil), *C. citratus* (lemon-grass oil) *E. globulus* (nilgiri oil), *A. indica* (neem oil) followed by *Mentha spicata* (mint oil) against *E. coli* and *A. niger* (Table 1).

The characterised AgNPs were then used for functionalisation with the antimicrobial plant oils. The emulsion appears brownish in colour. It was tested to determine its viscosity, pH and foam formation. The oil conjugated AgNPs were characterised by LM20, which revealed the size of AgNPs to be 85 nm (Fig. 3) with a zeta potential of -34.8 indicating the stability of the nanoparticles (Fig. 4).

The antimicrobial activities of plant oil, AgNPs and AgNPs functionalised with herbal oils were determined by the disc

Table 1 Antimicrobial activity of the oils of medicinal plants against *A. fumigatus* and *S. aureus* (ATCC 33591) growth in vitro

Sl. no.	Plant oils	<i>A. fumigatus</i> , µg				<i>S. aureus</i> , µg			
		100	50	25	12.5	100	50	25	12.5
1	<i>A. indica</i> (neem oil)	16 ± 0.5	13 ± 0.5	12 ± 1.1	10 ± 1.1	20 ± 0.5	14 ± 0.5	12 ± 0.5	10 ± 0.5
2	<i>C. martini</i> (tikhadi oil)	24 ± 0.5	22 ± 0.5	21 ± 0.5	19 ± 0.5	26 ± 0.5	20 ± 0.5	19 ± 0.5	15 ± 0.5
3	<i>C. citratus</i> (lemon-grass oil)	22 ± 0.5	20 ± 0.5	19 ± 0.5	18 ± 0.5	21 ± 0.5	20 ± 0.5	18 ± 0.5	18 ± 0.5
4	<i>E. globulus</i> (nilgiri oil)	22 ± 0.5	20 ± 0.5	19 ± 0.5	18 ± 0.5	24 ± 0.5	21 ± 0.5	20 ± 0.5	19 ± 0.5
5	<i>M. spicata</i> (mint oil)	16 ± 0.5	13 ± 0.5	12 ± 0.5	10 ± 0.5	18 ± 0.5	15 ± 0.5	12 ± 0.5	09 ± 0.5

diffusion method as a simple and fast method to distinguish the antimicrobial activity. Here, we observed that AgNPs functionalised with plant oil formulation show maximum zone of inhibition in comparison to AgNPs against *S. aureus*, *E. coli*, *A. niger* and *A. fumigatus*. The results indicated that the leaf oil and AgNPs alone exhibit less antimicrobial effect (Fig. 6) against *S. aureus*, *E. coli*, *A. niger* and *A. fumigatus*. However, AgNPs functionalised with antimicrobial oils showed the maximum antimicrobial activity against the above tested microorganisms (Fig. 6).

We developed soap, shampoo and ointment by using AgNPs functionalised with plant antimicrobial oils for animals. We also evaluated the properties and the viscosity of herbal shampoo which was found to be 1710 cP. The pH of herbal soap and shampoo was found to be 6.5 and 6.7, respectively. The spreadability of herbal soap and shampoo was found to be 46.4 and 53.1%, respectively. Table 2 shows the results of the wound healing effects of the ointment formulations. There was a general decrease in wound area on application of the ointments, with respect to time. From % wound contraction, it is concluded that the developed ointment produced greater wound contraction compared with the standard ointment and wound contraction increases as the day passes, proving significant wound healing activity. On the basis of the results obtained in the present investigation, it is possible to conclude that AgNPs functionalised with mixed herbal oils can be potentially used to develop effective antibacterial and antimicrobial shampoo and soap for veterinary use to reduce animal skin infections, with enhanced wound healing activity. The broad spectrum activity, effectiveness, lowcost and eco-friendly approach made it superior over the other rinse-off products (Fig. 7).

4 Discussion

The development of antimicrobial shampoo and soap for animals by using simple green biological methods could be a promising route to potentially environmentally friendly applications of nanoparticles. Nanoparticles, being nanosized, exhibit smaller volume to greater surface area ratio and have high dispersion property; hence, they are utilised for the formulation of nano-functionalised antimicrobial products.

For the nanoformulation of AgNPs functionalised with antimicrobial plant oils, the AgNPs were first biologically synthesised and characterised. On addition of aqueous silver ions, a brown colour developed, which when scanned in the range of 200–800 nm in a UV–vis spectrophotometer, showed a sharp peak at 415 nm with respect to surface plasmon resonance exhibited by AgNPs. These kinds of observations were previously reported by using different fungi (28, 29, 30). FTIR spectra confirm the presence of proteins secreted by the plant, in turn stabilising the synthesised AgNPs. Further characterisation with LM20 revealed the size of AgNPs alone to be 63 nm and that of AgNPs after conjugation to be 85 nm. Similarly, zeta potential measurement

revealed the zeta potential as -34.8 , proving the stability of nano-functionalised antimicrobial plant oils.

Since ancient times, silver and silver-based compounds have been known for their antimicrobial activity because of their antiseptic properties to several kinds of bacterium and fungi, including *E. coli* and *S. aureus*, *A. niger* and *A. fumigatus* [23–25]. Silver-based antimicrobial agents have received much attention, because of the low toxicity of the nanoparticles to human cells [26–28]. Several biological applications have been widely documented for its low concentration use [29]. AgNPs show excellent antibacterial properties and various research groups have investigated the mechanism behind the AgNPs-mediated antibacterial activity [3, 30–33]. As the size of the silver particles decreases to the nanoscale, their antibacterial efficacy increases because of their larger total surface area per unit volume [3, 30, 31] and hence due to the large surface area, it is used for the bioconjugation [34]. If the aim is to develop a general, simple (for example, single-step) procedure to make a solid surface bactericidal shampoo and soap for animal, then covalent attachment of oil with AgNPs is probably a viable option given the paucity of derivatisation amenable functional groups on most common surfaces. Hence, we aimed for the exploitation of the functionalised AgNPs to enhance the antimicrobial potential of plant oils. Thus, nanoparticles help in the development of potent antibacterial and antimicrobial shampoo and soap for animals, coatings in a single step at ambient conditions without using external reagents or excessive energy for practical applications.

The antimicrobial activity of AgNPs was reported in a series of studies [3, 35–37]. In the current investigation, plant oils conjugate with AgNPs were effective against *S. aureus*. Similar to these observations, Govindaraju *et al.* [37] showed that plant AgNPs formed zone of inhibition when the synthesised nanoparticles were tested against *Pseudomonas aeruginosa*, *S. aureus*, *Aspergillus flavus* and *A. niger*. A number of theories for the antimicrobial actions of colloidal silver solution have been proposed. For example, alteration of permeability of cell membrane [38], release of lipopolysaccharides and membrane proteins [39], generation of free radicals responsible for the damage of membrane [40] and dissipation of the proton motive force resulting in the collapse of the membrane potential [41]; however, the exact mechanism has not been fully understood [42]. The wound healing results of the formulated cream prove the superior activity of the AgNPs functionalised with plant oils. AgNPs were reported to heal wounds as there was better collagen alignment after healing, which results in mechanical strength [43]. The main aim is to achieve rapid recovery with minimal scarring. Experimental evaluation of the wound healing activity of AgNPs showed an increased wound contraction and epithelialisation in treated animals. Thus, it can be used to develop various products for animal health care.

5 Conclusion

In conclusion, the biosynthesised AgNPs using plant extract of *A. indica* showed potential antimicrobial activity against various bacterial and fungal pathogens present on skin of animals in combination with antimicrobial plants oils. This could be further used in the formulation of potential antimicrobial agents for veterinary dermatology. The current study revealed that AgNPs can be synthesised with a simple method using plant extract. The

Table 2 Percentage of wound healing contraction

Formulation code	4th day	8th day	12th day	16th day
control (wound)	21.9	36.62	36.62	36.62
standard streptomycin	37.1	60.3	60.1	60.1
test	47.1	61.8	67.7	86.7

TEM analysis showed that the size of the synthesised AgNPs ranged from 15 to 35 nm. AgNPs functionalised with mixed plant oils showed a higher antimicrobial activity compared with plant oil alone or AgNPs. AgNPs functionalised with plant oils help to maximise benefit for veterinary, pharmaceutical and biological products. Disease control in animals is multifaceted, and the more traditional emergency products are required for preventive measures. In professional hospitals and home, attention is being paid to veterinary needs. Antimicrobial products of plant origin are valuable, versatile and safe and have a crucial and specific role in controlling bacterial diseases in animals. As this approach is based on clean, eco-friendly and low-cost technology, we believe that it will be a good alternative therapy to solve the continuous antibiotic resistance developed by many bacterial pathogens, and will be utilised in various animal contacting areas in medicine.

6 Acknowledgment

The authors wish to express their thanks to DST-Nanomission for their financial support to carrying out this research work.

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